

Varun Tyagi

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[Portfolio](#) | [Linkedin](#) | [Github](#) | [LeetCode](#)

EDUCATION

Dehradun Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Dehradun, Uttarakhand

Aug. 2023 - May 2027

EXPERIENCE

Celebal Technologies

June 2026-August2026

Data Engineer Intern

Tech Stack: Python, SQL, Pandas, PySpark, Apache Spark, Azure Data Factory (ADF), Azure Blob Storage, Azure Databricks

- Completed structured training in Python, SQL, Apache Spark, Azure Data Factory, Azure Blob Storage, and Azure Databricks as part of the Data Engineering internship program.
- Built ETL pipelines using Azure Data Factory by integrating Blob Storage and SQL data sources, implementing Copy Data, Metadata, and pipeline orchestration activities.
- Performed data cleaning, transformation, aggregation, and exploratory analysis using Pandas and SQL, including handling missing values, duplicate removal, joins, CTEs, subqueries, and window functions.
- Developed Spark DataFrame transformations and implemented Delta Lake MERGE operations while working with Bronze-Silver-Gold data architecture and data validation workflows.

TECHNICAL SKILLS

- Languages:** Java, JavaScript, TypeScript, SQL, HTML/CSS
- Frontend:** React.js, Next.js, Tailwind CSS
- Backend:** Node.js, Express.js, RESTful API, JWT Authentication
- Database:** MongoDB, Mongoose
- Developer Tools:** Git, GitHub, VS Code, Postman
- Core CS:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks

PROJECTS

AlgoFlow — Interactive DSA Pattern Visualizer | [LIVE](#) | [GitHub](#)

Tech Stack: Next.js, React.js, TypeScript, Tailwind CSS

- Developed an interactive DSA visualization platform featuring **6 algorithm modules** with step-by-step execution playback.
- Created **25+ reusable React components** to support visualization workflows and UI consistency.
- Designed animation-driven visualizations for algorithm state transitions and execution tracking.
- Implemented **dual-mode comparison** to visualize brute-force and optimized approaches side-by-side with operation counts and time/space complexity insights for performance analysis.
- Built a modular architecture enabling rapid addition of new algorithm visualizers.

JobSync – AI-Powered Job Management Portal | [LIVE](#) | [GitHub](#)

Tech Stack: React.js, Node.js, Express.js, MongoDB, Tailwind CSS, JWT, Gemini AI

- Engineered a full-stack role-based job portal supporting Student and Recruiter workflows with secure JWT authentication, protected REST APIs, and CRUD operations across **15+ application screens**.
- Implemented an AI Career Assistant by integrating **Google Gemini API** to generate interview questions, create personalized cover letters, and analyze resumes with AI-driven skill recommendations, reducing manual preparation time by an estimated 70%.
- Designed a complete **Applicant Tracking System (ATS)** enabling recruiters to post jobs, review applicants, and update application status (Pending, Accepted, Rejected) with real-time dashboard synchronization and **role-based authorization**.
- Optimized user experience through advanced features including job search, filtering, saved jobs, application tracking, responsive dashboards, toast notifications, loading states, and mobile-first UI, delivering 100% responsive layouts across desktop and mobile devices.

ACHIEVEMENTS

- Solved **600+** Data Structures & Algorithms problems on LeetCode, achieving a peak contest rating of **1578**.
- Selected among 98 out of 1000+ students through the campus recruitment process for the Data Engineer Internship at Celebal Technologies.